The Informax Decoupler in ScaPre: Effects of Channel Weighting on Concept Removal and Preservation

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1 Background and Question

ScaPre (**Scalable and Precise Concept Unlearning**) is a closed-form method that edits cross-attention projections to remove selected concepts while preserving related content.

The **Informax Decoupler** is its channel-weighting component. It uses **mutual information (MI)**, which measures the dependence between a binarized channel activation and the target-versus-neutral label.

We ask whether these weights meaningfully distinguish channels and whether that distinction matters at image level.

2 From Activations to MI Weights

We follow activations → top-half assignment → binary MI → channel weights → suppression matrix.

For $W \in \mathbf{R}^{d_{\text{out}} \times d_{\text{in}}}$, channel i is row $W_{i:}$. Given $N = 2n$ labeled inputs, an input feature s produces

$$a_i(s) = W_{i:}s, \qquad z = \mathbf{1}\{a_i(s) > \tau_i\} \in \{0, 1\},$$

where the analyzed implementation sets τ_i to the median channel activation. The label $y \in \{0, 1\}$ distinguishes target ($y = 1$) from neutral ($y = 0$) inputs. With joint counts n_{zy}, let $p_i(z, y) = n_{zy}/N$; the marginals $p_i(z)$ and $p_i(y)$ are obtained by summing the corresponding joint probabilities. The channel score is

$$MI_i = \sum_{z\in\{0,1\}} \sum_{y\in\{0,1\}} p_i(z, y) \log \frac{p_i(z, y)}{p_i(z)p_i(y)}.$$

For target concept c_ℓ, this gives $MI_i^{(\ell)}$. ScaPre then defines

$$MI_i = \max_\ell MI_i^{(\ell)}, \quad \alpha_i = \frac{MI_i}{\max_j MI_j}, \quad B = \operatorname{diag}(\alpha).$$

The objective contains $\operatorname{tr}(W^\top BW)$, so channel-specific weighting requires meaningful differences among the MI_i values.

3 Why Binary MI Has Few Values

Consider n target and n neutral samples. Without threshold ties, the median split places n samples in each binary state. If k target samples are in the upper half, the table is

	target	neutral
$z = 1$	k	$n - k$
$z = 0$	$n - k$	k

Balanced sampling fixes the column totals, while the median split fixes the row totals. Thus MI is determined only by k:

$$\begin{aligned} MI(k) &= 2\frac{k}{2n} \log \frac{k/(2n)}{(1/2)(1/2)} + 2\frac{n-k}{2n} \log \frac{(n-k)/(2n)}{(1/2)(1/2)} \\ &= \frac{k}{n} \log \frac{2k}{n} + \frac{n-k}{n} \log \frac{2(n-k)}{n} = MI(n-k). \end{aligned}$$

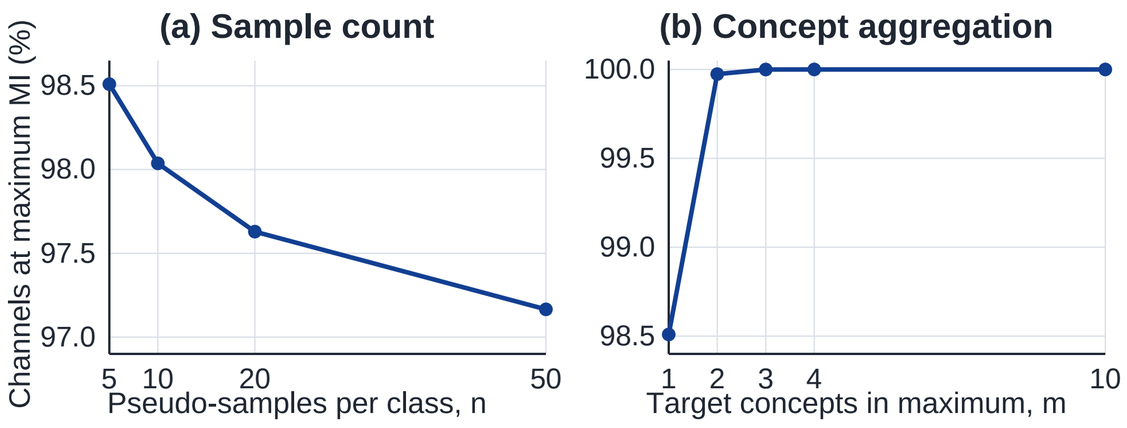
The non-redundant cases are $k = 0, 1, \ldots, \lfloor n/2 \rfloor$, giving exactly

$$\boxed{\left\lfloor \frac{n}{2} \right\rfloor + 1}$$

distinct numerical values under the no-tie condition. For $n = 5$, the no-tie case yields exactly three values: 0.0201, 0.1927, and $\ln 2$.

4 Empirical Saturation and Aggregation

The repository uses the empty-string embedding as the neutral concept; the measurements below follow that protocol.



Panel (a) increases the pseudo-sample count from $5 + 5$ to $50 + 50$, reducing the maximum-MI fraction only from 98.52% to 97.17%. Panel (b) averages all target subsets of size m across five seeds; every observed subset reaches 100% saturation from $m = 4$ onward.

5 Paper and Repository Weighting

For m target concepts, the paper defines the aggregate as

$$MI_i^{(m)} = \max_{1\le\ell\le m} MI_i^{(\ell)}.$$

In the measured ten-target data, every channel has reached $MI_i^{(m)} = \ln 2$ by $m = 4$. The paper's normalization therefore gives

$$\alpha_i = \frac{\ln 2}{\ln 2} = 1, \qquad B = I, \qquad \operatorname{tr}(W^\top BW) = \|W\|_F^2.$$

The repository instead applies, for each concept,

$$MI \to z\text{-score} \to [\text{sigmoid}(\cdot/0.7)]^8 \to \max_\ell .$$

The repository retains small channel-to-channel differences, so we next test whether their assignments matter.

6 Final Channel Controls

We evaluate Sections 6-7 on Stable Diffusion v1.5 with reconstructed ImageNet-Confuse5: 10 target concepts, 15 similar non-target concepts, and 120 images per concept and seed. Both tables report means over five fixed edit seeds.

The metrics are residual target accuracy $U \downarrow$, preserve accuracy $P \uparrow$, and

$$O = \frac{2(100 - U)P}{(100 - U) + P} \uparrow .$$

Constant mean and Identity B change only the final α.

Variant	$U \downarrow$	$P \uparrow$	$O \uparrow$
Repository baseline	19.27	40.46	53.90
Constant mean	19.18	**40.81**	**54.23**
Identity B (different scale)	19.23	40.68	54.11

The controls that modify only the final weights remain near the baseline.

7 Replacing Binary MI with Cosine Alignment

Because binary MI saturates, we test a continuous channel score in the earlier per-concept mask; the final repository weights stay unchanged.

Let $w_i = W_{i:}$ and $d_\ell = c_\ell - c_0$, where c_0 is the empty-string vector. A continuous alignment score is

$$r_i^{(\ell)} = \frac{(w_i^\top d_\ell)^2}{(\|w_i\|_2^2 + \varepsilon)(\|d_\ell\|_2^2 + \varepsilon)} \approx \cos^2 \theta_i.$$

We test two replacements:

$$\text{Transformed cosine: } \alpha^{(\ell)} = g(r^{(\ell)}),\ g(r) = [\text{sigmoid}(\text{z-score}(r)/0.7)]^8,$$
$$\text{Direct: } \alpha^{(\ell)} = r^{(\ell)}.$$

In g, z-scoring is across channels; the remaining operations are elementwise.

Variant	$U \downarrow$	$P \uparrow$	$O \uparrow$
Repository baseline	19.27	40.46	53.90
Transformed $\cos^2$	19.77	**45.84**	**58.35**
Direct $\cos^2$	**19.10**	40.51	53.99

Transformed $\cos^2$ improves P and O, but largely fails to erase tiger cat: U rises by 40.17 points on average. Direct $\cos^2$ avoids this failure but remains close to the baseline.

8 Conclusion

- Binary MI barely distinguishes channels: saturation remains at 97.17% with $50 + 50$ pseudo-samples, and the paper's aggregation gives $B = I$ from four targets onward.
- Changing only the final α has little effect; the repository's small channel differences show no clear image-level advantage.
- Cosine alignment changes the erase–preserve trade-off, but does not provide a reliable replacement.

Scope: SD v1.5, reconstructed Confuse5, and five fixed edit seeds.

Reference

K. Deng et al., "Forget Many, Forget Right: Scalable and Precise Concept Unlearning in Diffusion Models," *ICLR*, 2026.